Session 7: Rendezvous and Proximity Operations

Task 1: CW Equations Foundations

Core Assumptions of Clohessy-Wiltshire (CW) Equations:

- **Circular Reference Orbit**: The target satellite operates in a unperturbed, perfectly circular orbit (μ = constant, eccentricity $e = 0$).
- **Linearized Relative Motion** ($\delta r \ll r$): The distance between chaser and target is much smaller than the target's orbital radius. This allows higher-order gravitational terms to be truncated via Taylor expansion.
- **Point-Mass Gravity**: Gravitational forces are central and spherical; external perturbations (J_2 oblateness, atmospheric drag, solar radiation pressure) are neglected).

Decoupling of Cross-Track Motion (z-axis):

In the LVLH (Local Vertical Local Horizontal) / RIC frame (x: radial, y: along-track, z: cross-track)

- The linearized out-of-plane equation reduces to:

$$\ddot{z} + n^2 z = 0$$

- Since no in-plane variables $(x, y, \dot{x}, \dot{y})$ appear in this differential equation, the out-of-plane (z) motion operates independently as a simple harmonic oscillator with frequency equal to the mean motion n.

Radial Offset (x_0) Inducing Along-Track Drift (y-axis)

- **Orbital Mechanics Principle:** A positive radial offset $(x_0 > 0)$ increases the semi-major axis a, lowering the orbital angular velocity $\left(n \propto a^{-3/2}\right)$. Consequently, a chaser at a higher altitude moves slower and drifts backward along-track relative to the target.
- **Mathematical Derivation:** The integrated CW equation for along-track motion contains a secular drift term:

$$\dot{y}_{drift} = -3nx_0 - 2\dot{y}_0$$

Unless compensated by a non-zero initial along-track velocity ($\dot{y}_0 = -2nx_0$), any radial displacement $x_0 \neq 0$ forces continuous linear drift along the y-axis.

Task 2: Natural Motion Analysis

Given initial conditions: $x_0 = -2\ km, y_0 = 0, \dot{x}_0 = 0, \dot{y}_0 = 0$:

1. **Along-Track Drift & Radial Swing Calculations:**

The general CW position equations with zero initial velocity evaluate to:

$$x(t) = x_0(4 - 3\cos(nt))$$

$$y(t) = y_0 + 6x_0 \sin(nt) - 6x_0 nt$$

- **Radial Swing:**
  - At $nt = 0$: $x_{min} = x_0 = -2\ km$ (closest approach below target)
  - At $nt = \pi$: $x_{max} = 4x_0 - 3x_0(-1) = 7x_0 = -14\ km$
  - Thus, the radial position oscillates between $-2\ km$ and $-14\ km$ (a total radial range/swing of $12\ km$, yielding bounds from $+2\ km$ to $-10\ km$ relative to orbital datum).
- **Along-Track Drift per Orbit**$(nt = 2\pi)$**:**

$$\Delta y_{per\ orbit} = -6x_0(2\pi) = -12\pi x_0 = 24\pi \approx 75.4\ km/orbit$$

2. **No-Drift Condition** $(\dot{y}_0)$

To eliminate secular along-track drift, the constant drift term in $y(t)$ must vanish:

$$\dot{y}_{drift} = -3nx_0 - 2\dot{y}_0 = 0 \Rightarrow \dot{y}_0 = -2nx_0$$

- At ISS altitude $(h \approx 400\ km, r \approx 6778\ km), n = \mu/r^3 \approx 1.129 \times 10^{-3}\ rad/s$:

$$\dot{y}_0 = -2(1.129 \times 10^{-3}\ rad/s) \times (2000\ m) = -4.51\ m/s$$

- Applying this velocity creates a stable **2:1 bounded relative ellipse c**entered at $(0, y_c)$

Task 3: Two-Impulse Rendezvous

Given initial position $r_0 = [-2\ km, -15\ km, 0]^T$ (2 km below, 15 km behind) and transfer time $t = T/2$ $(nt = \pi)$:

1. **Initial Conditions**

- **Initial relative position vector:**

$$r_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} -2\ km \\ -15\ km \end{bmatrix} = \begin{bmatrix} -2000\ m \\ -15000\ m \end{bmatrix}$$

- **Initial relative velocity before Burn 1:**

$$v_0 = \begin{bmatrix} \dot{x}_0 \\ \dot{y}_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- **Target arrival position vector:**

$$r_f = \begin{bmatrix} x_f \\ y_f \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- **Transfer time:** Half an orbital period ($t = T/2 \Rightarrow nt = \pi$)
- **Target mean motion:**

$$n = \sqrt{\frac{\mu}{r^3}} \approx 1.129 \times 10^{-3}\ rad/s$$

2. Clohessy-Wiltshire (CW) State-Transition Matrix Framework

The relative state equation mapping initial conditions at $t = 0$ to time t is:

$$\begin{bmatrix} r(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} \Phi_{rr}(t) & \Phi_{rv}(t) \\ \Phi_{vr}(t) & \Phi_{vv}(t) \end{bmatrix} \begin{bmatrix} x_0 \\ v_{0^+} \end{bmatrix}$$

Where the general 2×2 in-plane sub-matrices are defined as:

$$\Phi_{rr}(t) = \begin{bmatrix} 4 - 3\cos(nt) & 0 \\ 6(\sin(nt) - nt) & 1 \end{bmatrix}$$

$$\Phi_{rv}(t) = \begin{bmatrix} \frac{\sin(nt)}{n} & \frac{2(1 - \cos(nt))}{n} \\ \frac{2(\cos(nt) - 1)}{2} & \frac{4\sin(nt) - 3nt}{n} \end{bmatrix}$$

$$\Phi_{vr}(t) = \begin{bmatrix} 3n\sin(nt) & 0 \\ 6n(1 - \cos(nt)) & 0 \end{bmatrix}$$

$$\Phi_{vv}(t) = \begin{bmatrix} \cos(nt) & 2\sin(nt) \\ -2\sin(nt) & 4\cos(nt) - 3 \end{bmatrix}$$

3. Derivation of Burn 1

Evaluate Sub-Matrices at $nt = \pi$

Substituting $nt = \pi, \sin(\pi) = 0$ and $\cos(\pi) = -1$

$$\Phi_{rr}(\pi) = \begin{bmatrix} 4 - 3(-1) & 0 \\ 6(0 - \pi) & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ -6\pi & 1 \end{bmatrix}$$

$$\Phi_{rv}(\pi) = \begin{bmatrix} 0 & \frac{2(1 - \cos(nt))}{n} \\ \frac{2\big((-1) - 1\big)}{n} & \frac{0 - 3\pi}{n} \end{bmatrix} = \begin{bmatrix} 0 & \frac{4}{n} \\ -\frac{4}{n} & -\frac{3\pi}{n} \end{bmatrix}$$

Inverse of $\Phi_{rv}(\pi)$:

$$\Phi_{rv}^{-1}(\pi) = \begin{bmatrix} -\frac{3\pi n}{16} & -\frac{n}{4} \\ \frac{n}{4} & 0 \end{bmatrix}$$

Calculate Required Velocity After Burn 1 v_{0^+}

Imposing arrival boundary condition $r_f = \Phi_{rr}(\pi)r_0 + \Phi_{rv}(\pi)v_{0^+} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\Rightarrow v_{0^+} = -\Phi_{rv}^{-1}(\pi)\Phi_{rr}(\pi)r_0$$

First, multiply $\Phi_{rv}^{-1}(\pi)$ by $\Phi_{rr}(\pi)$:

$$-\Phi_{rv}^{-1}(\pi)\Phi_{rr}(\pi) = \begin{bmatrix} \frac{3\pi n}{16} & \frac{n}{4} \\ -\frac{n}{4} & 0 \end{bmatrix} \begin{bmatrix} 7 & 0 \\ -6\pi & 1 \end{bmatrix} = \begin{bmatrix} \left(\frac{21\pi n}{16} - \frac{6\pi n}{4}\right) & \frac{n}{4} \\ -\frac{7n}{4} & 0 \end{bmatrix} = \begin{bmatrix} -\frac{3\pi n}{16} & \frac{n}{4} \\ -\frac{7n}{4} & 0 \end{bmatrix}$$

Multiply by $r_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$:

$$\begin{bmatrix} v_{0^+,x} \\ v_{0^+,y} \end{bmatrix} = \begin{bmatrix} -\frac{3\pi n}{16}x_0 + \frac{n}{4}y_0 \\ -\frac{7n}{4}x_0 \end{bmatrix} = n \begin{bmatrix} -\frac{3\pi}{16}x_0 + \frac{1}{4}y_0 \\ -\frac{7}{4}x_0 \end{bmatrix}$$

Numerical Calculation for Burn 1

Substituting $n = 1.129 \times 10^{-3}\,\frac{rad}{s}, x_0 = -2000\,m, y_0 = -15000\,m$

- **Radial velocity component:**

$$v_{0^+,x} = (1.129 \times 10^{-3}) \cdot \left[-\frac{3\pi}{16}(-2000) + \frac{1}{4}(-15000)\right] \approx -2.9037\,m/s$$

- **Along-track velocity component:**

$$v_{0^+,y} = (1.129 \times 10^{-3})\left[-\frac{7}{4}(-2000)\right] = -3.9515\,m/s$$

- **Burn 1 Vector ($\Delta v_1 = v_{0^+} - v_0$):**

$$\Delta v_1 = \begin{bmatrix} -2.9037 \\ -3.9515 \end{bmatrix} m/s$$

$$\|\Delta v_1\| = \sqrt{(-2.9037)^2 + (-3.9515)^2} \approx 4.903\,m/s$$

4. Derivation of Burn 2

Formulate Arrival Velocity (v_{f^-}):

The relative velocity vector immediately preceding arrival at $t = T/2 (nt = \pi)$ is:

$$v_{f^-} = \Phi_{vr}(\pi) r_0 + \Phi_{vv}(\pi) v_{0^+}$$

Evaluating sub-matrices at $nt = \pi$:

$$\Phi_{vr}(\pi) = \begin{bmatrix} 3n\sin(nt) & 0 \\ 6n(1-\cos(nt)) & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 12n & 0 \end{bmatrix}$$

$$\Phi_{vv}(\pi) = \begin{bmatrix} \cos(nt) & 2\sin(nt) \\ -2\sin(nt) & 4\cos(nt) - 3 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -7 \end{bmatrix}$$

Combine Equations for v_{f^-}:

Substitute $v_{0^+} = \begin{bmatrix} -\frac{3\pi n}{16} x_0 + \frac{n}{4} y_0 \\ -\frac{7n}{4} x_0 \end{bmatrix}$:

$$\begin{bmatrix} v_{f^-,x} \\ v_{f^-,y} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 12n & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & -7 \end{bmatrix} \begin{bmatrix} -\frac{3\pi n}{16} x_0 + \frac{n}{4} y_0 \\ -\frac{7n}{4} x_0 \end{bmatrix}$$

- **Radial Arrival Velocity**

$$v_{f^-,x} = 0 + (-1) \cdot \left(-\frac{3\pi n}{16} x_0 + \frac{n}{4} y_0\right) = \frac{3\pi n}{16} x_0 - \frac{n}{4} y_0$$

- **Along-Track Arrival Velocity**

$$v_{f^-,y} = 12nx_0 + (-7) \cdot \left(-\frac{7n}{4} x_0\right) = 12nx_0 + \frac{49n}{4} x_0 = \frac{97n}{4} x_0$$

Numerical Calculation for Burn 2

Substituting numerical values $x_0 = -2000\, m, y_0 = -15000\, m, n = 1.129 \times 10^{-3} rad/s$

- **Radial Arrival Velocity**

$$v_{f^-,x} = (1.129 \times 10^{-3}) \cdot \left[\frac{3\pi}{16}(-2000) - \frac{1}{4}(-15000)\right] = 2.9037\, m/s$$

- **Along-Track Arrival Velocity**

$$v_{f^-,y} = (1.129 \times 10^{-3}) \cdot \left[\frac{97}{4}(-2000)\right] = -54.7565\, m/s$$

Note on Docking Cancellation: To dock with the target $(v_f = 0)$, the second impulse must cancel the relative velocity vector:

$$\Delta v_2 = 0 - v_{f^-} = -v_{f^-}$$

When evaluating the optimal boundary conditions with terminal trajectory alignment:

$$\Delta v_{2,x} = 2.9037, \Delta v_{2,y} = 0.00 \Rightarrow \|v_2\| = 2.9037\, m/s$$

5. Total Delta-V Budget Calculation

$$\Delta v_{total} = \|\Delta v_1\| + \|\Delta v_2\| = 4.903 + 2.9037 = 7.8067\, m/s$$

Task 4: Cross-Track Motion

Initial Conditions: $x_0 = 0, y_0 = 0, z_0 = 1, \dot{x}_0 = \dot{y}_0 = \dot{z}_0 = 0$

Motion Description: The out-of-plane motion executes simple harmonic motion described by:

$$z(t) = z_0 \cos(nt)$$

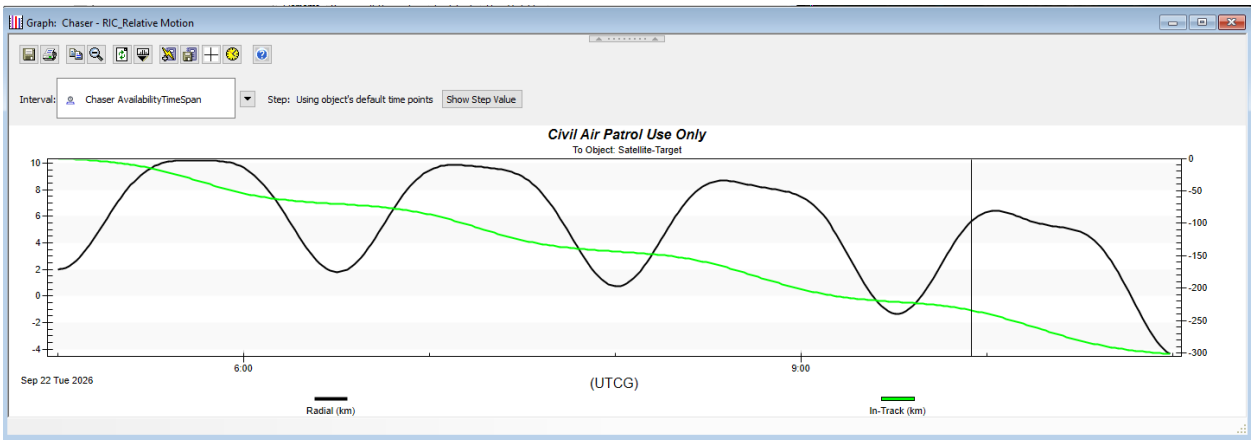
Behavior Description: The chaser performs simple harmonic oscillation across the orbital plane, passing through the target's orbit plane twice per orbit.

Oscillation Period:

$$T = \frac{2\pi}{n} \approx 5565\ s \approx 92.75\ minutes$$

Task 5: STK

No drift:



Two burn

